

Scratch Pen Tool Intro

Let's Learn to Draw with Code!

Grade 2 Lesson Guide



What is Scratch?

Scratch is a fun coding platform where you can create stories, games, and amazing art! No typing is needed because Scratch uses colorful blocks that snap together like puzzle pieces.

- Make your own animations and games.
- Draw incredible pictures using smart code.
- Share your creations with friends and family.



Meet the Pen Tool

The Pen Tool is an extension in Scratch that lets your character (sprite) draw lines and shapes as it moves around the screen—just like a pencil on paper!

- **Pen Down:** Lowers the pencil onto the stage to start drawing.
- **Pen Up:** Lifts the pencil up so you can move around without leaving lines.
- **Change Pen Color & Size:** Pick your favorite colors and make your lines thick or thin!



Essential Pen Blocks

Here are the primary blocks you will use to build your digital artwork:

pen down — Starts drawing trails.

pen up — Stops drawing trails.

set pen color to [🎨] — Picks your color choice.

change pen size by (1) — Makes lines thicker or thinner.

clear all — Erases all artwork on screen to start over clean.

stamp — Leaves a stamped copy of your sprite image on the screen!



Getting Started in 3 Easy Steps

Step 1: Add the Pen Extension

Open a new project at scratch.mit.edu. Click the blue **Extensions button** at the very bottom left corner of your screen, find the **Pen extension**, and click it to unlock the green blocks!

Step 2: Ready Your Pencil

Drag out your yellow start block and snap the green **pen down** block right under it:

when green flag clicked

pen down

Step 3: Move and Watch the Magic!

Add a blue motion block like **move (100) steps** underneath. Click the green flag and watch your sprite leave a colorful trail behind it!



Mini Project: Draw a House!

Ready for a creative challenge? Let's use code blocks to build a house:

1. **The Base:** Put the pen down, move, and turn 4 times to draw a square base.
2. **The Roof:** Move to the top and draw a 3-sided triangle roof.
3. **Details:** Use **pen up** to change positions to draw windows and a door!
4. **Decorate:** Change pen colors to draw a bright yellow sun, green grass, or fluffy clouds!



Let's Try a Quick Quiz!

Q1: What block makes your sprite START drawing?

A) Pen Up **B) Pen Down ✓** C) Move

Q2: Which block STOPS the sprite from drawing?

A) Pen Down B) Set Pen Color **C) Pen Up ✓**

Q3: Which block changes the line color?

A) Set Pen Color ✓ B) Pen Down C) Move



Summary: What We Learned Today

- Scratch makes learning code an exciting puzzle adventure!
- The Pen Tool draws an automatic line path as your sprite moves.
- Always combine **Pen Down** to draw, **Pen Up** to move freely, and **Clear All** to try again.
- Loops make complex drawings (like circles and spirals) simple and easy.

Keep exploring, stay creative, and have fun coding! 🎈